# Rutika Avinash Kadam

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## SUMMARY

Data Scientist with 4 years of experience analyzing large-scale datasets, developing predictive models, and translating complex data into actionable operational decisions. Skilled in Python, SQL, statistical modeling, NLP, and multimodal GenAI, with experience in model evaluation, experimentation, dashboarding, and cloud-based ML workflows.

## EXPERIENCE

**Data Scientist (Graduate Researcher), Stony Brook Medicine | USA** **Jul 2025 – Present**

- Built an end-to-end multimodal RAG pipeline over 6K+ public-health document pages, using Azure Document Intelligence for layout-aware text and table extraction workflows.
- Generated GPT-4o figure captions and applied adaptive semantic chunking, organizing text, tables, and images into 27K+ retrieval-ready chunks for multimodal search experiments.
- Benchmarked FLAT, HNSW, and IVF_FLAT indexes in Milvus using p50/p95 latency and Recall@5; approximate indexes matched FLAT's Recall@5 while reducing latency by approximately 60%.
- Compared dense retrieval, MMR reranking, and hybrid vector-plus-BM25 retrieval with reciprocal rank fusion against synthetic ground truth, achieving 0.90 Recall@5 across modalities.
- Evaluated citation-grounded answers with RAGAS, scoring 0.82 faithfulness, 0.82 answer relevancy, 0.93 context precision, and 0.74 context recall across evaluated test cases.

**Data Scientist, Tata Consultancy Services | India** **Aug 2020 – Apr 2024**

- Queried 1M+ Qualys VMDR vulnerability records using SQL and Python EDA, identifying anomalies, threat vectors, and remediation priorities with the Vulnerability Management team.
- Built linear and logistic regression models on 150K findings to predict CVSS severity and exploit availability; exploit-availability prediction achieved 0.74 ROC-AUC, with backtested risk exposure 22% lower than FIFO.
- Clustered 150K+ vulnerability records using E5 embeddings and HDBSCAN, then fine-tuned DistilBERT to reproduce cluster-derived labels with 100% held-out agreement.
- Built Power BI dashboards tracking CVSS, exploit exposure, MTTR, and SLA compliance; post-implementation Jira ticket volume was 64% below the pre-dashboard baseline.
- Automated enterprise remediation workflows with SCCM and PowerShell, reaching 99% patch compliance and a 25% productivity improvement versus the pre-automation baseline across vulnerability operations.

**Data Intern, Zensar Technologies | India** **May 2019 – Jul 2019**

- Built incremental Azure Data Factory pipelines to load multichannel website traffic and operational data into Azure Data Lake Storage Gen2 for Swiggy India analytics.
- Developed Power BI dashboards for conversion funnels and marketing-channel attribution, identifying high-performing traffic sources and increasing high-intent traffic by 15% versus the prior period.

## PROJECTS

**Electric Vehicle Battery Capacity Prediction**

- Built an EV battery-capacity prediction model using routine charging data to address sessions lacking conditions for direct measurement; those conditions were present in only 56% of sessions, across 111K charging windows from 30 vehicles.
- Across 10 experiments with vehicle-wise five-fold cross-validation, the linear odometer baseline achieved 1.10 Ah RMSE versus 1.69 Ah for LSTM and 1.71 Ah for Transformer; ablation showed the 21-minute window constrained performance.
- Versioned data with DVC and tracked experiments in MLflow; containerized a FastAPI service with Docker, pushed the image to Azure Container Registry, and deployed it to Azure Container Apps, with CI running 43 tests per push.

**Airbnb Price Prediction: Seattle**

- Analyzed 3.8K Seattle Airbnb listings and 84K reviews using Python/NLTK sentiment analysis; room type and bedroom count were the strongest price predictors, while review sentiment showed no measurable relationship with price.
- Engineered 19 amenity features and benchmarked six linear, bagging, and boosting models using leakage-free 10-fold cross-validation, achieving $R^2 \approx 0.51$ and RMSE ≈ $62; boosting overfit on held-out folds.

## EDUCATION

**Stony Brook University, USA** - *Master of Science, Data Science* **Aug 2024 – May 2026**
**Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Large Language Models, Agentic AI, Big Data

**Pune University, India** - *Bachelor of Engineering, Information Łechnology* **Aug 2016 – May 2020**
**Coursework:** Statistics, Discrete Structures, Data Structures & Algorithms, System Design, Cloud Computing

## SKILLS

**Programming & Querying** : Python, SQL (MySQL, PostgreSQL, PL/SQL), R

**Data Analysis & Visualization** : Pandas, NumPy, Power BI, DAX, Tableau

**Statistical Analysis & Model Evaluation** : Exploratory Data Analysis (EDA), regression, classification, feature engineering, clustering, cross-validation, model benchmarking, backtesting, ablation studies

**Machine Learning & NLP** : scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers

**GenAI & Retrieval** : Large Language Models (LLMs), GPT-4o, multimodal RAG, LangChain, LangGraph, vector search (Milvus), hybrid retrieval, RAGAS

**Cloud & MLOps** : Azure Data Factory, Azure Data Lake Storage Gen2, Azure Databricks, Azure Container Apps, FastAPI, Docker, MLflow, DVC, Git